

Summary | Current clinical trials have high risk in terms of safety and efficacy because they are founded on cell-based and animal studies that fail to translate 90% of time in human trials.^{1,2} There are multi-billion dollar investments into making better models of disease, or using AI for drug discovery. We propose a third way that could avoid both of these proxies altogether. We propose a safe, 'direct-to-human' approach, where we simultaneously screen large drug libraries (1E3–1E7) whose action is restricted to a microscopic volume of living patient tissue. That will allow us to find exactly what cures the human disease, while also reducing systemic risks of clinical trials. If successful, we would revolutionize medicine: reduce or eliminate the dependency on animal models, accelerate clinical development, and reduce the costs of drugs. Lastly, with this high throughput, we could start treating emerging, or previously unsolved diseases immediately, avoiding years to decades of mechanistic discovery and target identification study. To catalyze this revolution, we propose Agnostic Screening for Accelerated Pharmaceuticals (**ASAP**) program, which offers:

- **Safety via insulation:** We restrict the drug exposure to just a few cubic millimeters of tissue, instead of the whole body, thus reducing systemic risk, making clinical trials safer at earlier stage.
- **Massively parallel testing:** Within that tiny volume, we screen up to 1E3-1E7 drugs simultaneously, preventing their diffusion away from the site. For example, in one project each cell is its own drug factory producing a different drug inside. Thus, each cell turns into an independent clinical drug trial. We then check which cells are cured and identify which drug was responsible. As a result in one step, and a short few-week procedure we obtain large numbers of drugs that already worked in human disease, while also avoiding systemic exposure to tested drugs.
- **Mechanistic understanding of real disease:** ASAP enables a massive-scale perturbation testing of real disease without AI, animal models, or other proxies. However, lessons from massive cell-level perturbation in human disease will also help identify more translatable drug targets, and train AI models better. Because the AI models train on existing data, often from animals, they may end up treating animals... on which they were trained. ASAP could feed more relevant data to AI.

ASAP we could find cures for disease we do not even understand leveraging high-throughput screening *in vivo*. The technologies for this paradigm shift are available, and we present one example of how it could be done. The regulatory pathway is also already paved for this concept.

The Problem | Why are so many diseases cured in mice, but not in real patients? Currently, drug development starts with identification of a **molecular target**, engagement of which in animals, organs on chips, or tissue culture warrants clinical testing. Eventually, these drugs fail in clinical trials approximately 90% of the time, after ~9 years in development, and having consumed >\$1B for just one successful drug^{1,2}. This classical drug discovery pipeline makes several assumptions: **1)** that the drugs that interact with the target actually treat the disease, and that the target is consequential not just associated with pathology; **2)** that a single target can be curative, and **3)** that what works in artificial models will also work in humans.

In ~10% cases these assumptions hold, and the drug succeeds. However, usually: 1) artificial models cannot replicate the lifetime that led to disease. For example, we do not wait ~60 years to naturally degrade dopaminergic neurons in rats – a hallmark of human Parkinson's Disease (PD); instead these neurons are killed off in mice with chemicals in weeks³. 2) The disease may not always driven by just one target, when biology typically relies on interactions of the networks of biomolecules⁴. Current pipeline is not well suited for multi-target trials⁵. 3) It is not clear if and when the new proxies will be better. E.g. we will only know whether organoids or AI led to cures, after the cures are achieved (or not).

The opportunity and Impact Outcomes | From the perspective of pure logic, the only way to ensure that we are treating real human disease, is to avoid proxies and test the drugs in patients. However, with the current technology, such testing would be dangerous, unethical, and economically unviable. A true paradigm shift must bypass mechanistic assumptions, proxy models, and need for decades of basic research. ASAP delivers this capability: a moonshot to safely test up to millions of drugs simultaneously

in vivo. At such scale, we can identify functional cures regardless of prior mechanistic understanding. If ASAP succeeds, we will reduce research timelines, development costs, and clinical failure rates for any cell-level disease. We estimate that targeting just 4 relevant conditions (vitiligo, NASH, ALS, progressive idiopathic lung fibrosis) has a \$22B addressable annual market. We also found further 27 highly relevant and feasible indications. Our program yields a >1,000-fold ROI on the first four targets alone.

The solution | There are many ways for ASAP to work. A specific example below is just one of the technologies that would form this broad program. Making large libraries of gene therapy vectors is now highly scalable and cost-effective (e.g., we made 1.3 billion variants for ~\$4,000⁶). By leveraging this capability, we turn few cubic mm of patient's own diseased tissue into a massively parallel drug screening facility through a simple, five-step process (**Fig. 1**):

1. Delivery: We inject this diverse vector library into a microscopic site of diseased tissue. By administering a low and highly localized dose, we ensure that approximately one vector enters each cell⁷. Consequently, each cell becomes its own "drug factory" with most expressing just a single, unique peptide drug. The vectors stay local and do not diffuse beyond a few mm radius⁸.

2. In Vivo Incubation: The peptide drugs are expressed locally for several weeks, interacting directly with the diseased cell's milieu.

3. Biopsy & Phenotypic Selection: We biopsy the injection site - safely removing the vectors from the body - and isolate the cells that show a phenotypic cure (e.g., the restoration of pigmentation in vitiligo).

4. Sequencing & Deconvolution: We sequence the vector DNA from these "cured" cells to identify the exact peptide responsible for the cure. If efficacy is low, we optimize the peptide via directed evolution.

5. Translation: We now have a human-vetted drug candidate. For rapidly progressing or ultra-rare diseases, this gene-encoded drug could be administered locally for an immediate cure using the same vectors as for the library. For broader applications, we push the candidate through the standard FDA pipeline with an unprecedented advantage: **we already know it was efficacious in the human disease** and safe in the tested region. Efficacy is responsible for majority of clinical trial failures⁹, and new tools

The Built-In Safety Testing: Because MAPS occurs at the single-cell level, we can also evaluate each treated cell for toxicity markers. Thus, each drug is pre-screened for both human efficacy and cellular safety. Animal studies usually (~70%¹⁰) predict toxicity in humans, with remaining cases possibly due to human immune system action¹¹ - something which would lead to the death of the cell expressing peptide candidate, preventing it from showing up in MAPS screen in the first place.

In this example, we target a skin disease, which is easy to biopsy, has an obvious phenotype, and needs a simple subcutaneous delivery with low systemic exposure to the drug. However, to make ASAP more broadly applicable – to treat ALS, MS, or systemic disorders, we will need to develop five areas:

1. Library design: Computational and experimental methods to determine which drugs should be included in the library. Milestone e.g.: Perform an in vivo screen which identifies drugs that are both efficacious and non-toxic in an animal model, to proof the screening technology.
2. In vivo screening methods to measure efficacy and toxicity of drugs: We will design the methodologies for screening drugs in vivo. What is the maximum number of drugs that can be tested at once? How to detect toxicity and efficacy on the cellular level, both locally and systemically? Milestone e.g.: determine max. random library size to identify a known therapeutic.

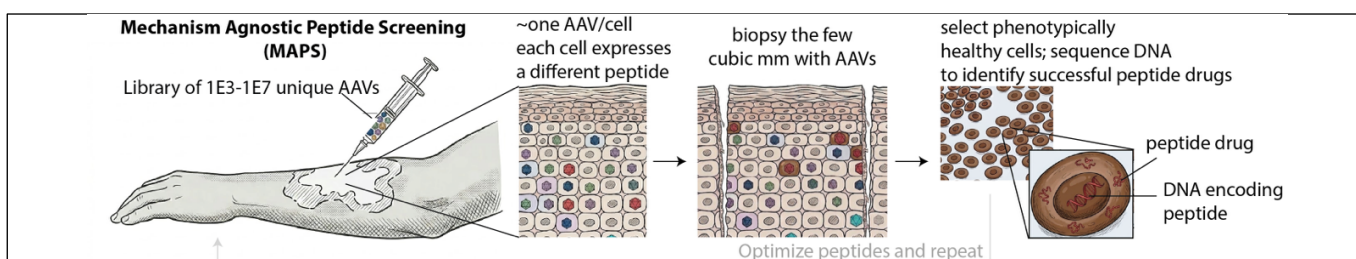


Figure 1 Mechanism Agnostic Peptide Screening concept - example. One of the many technologies that could be enabled by the ASAP program. Other technologies would address library design, delivery methods, noninvasive monitoring of drug action, and application to various other conditions. Many of the relevant technologies exist and can be integrated to enable ASAP.

3. Delivery methods for libraries and drug candidates: How to deliver the libraries to the right cells, and drugs to the right tissues? Which vectors to use for peptide screening, and what other technologies can be used for small molecule versions of MAPS? Milestone e.g.: No detectable library outside of injection site; identified peptide drugs stay within injection site in a skin disease.
4. Noninvasive methods of monitoring cellular disease phenotypes: Skin biopsy is simple, while for other organs, such as liver, or lung it is possible but undesired. Other organs, such as the spinal cord, are not routinely biopsied. We will leverage our synthetic serum markers, or noninvasive imaging, to monitor the drug success noninvasively, without biopsy. Milestone e.g.: detect >100 drugs with a blood test using Released Markers of Activity¹²⁻¹⁵ or other noninvasive methods.
5. Mechanism discovery and enabling the AI training on real disease data: we will develop methods to analyze the cellular phenotype and transcriptome after multiplexed drug testing to unravel a trove of data for understanding of disease, training AI models, and identifying consequential drug targets. Milestone e.g.: Identify the target for a known drug from a peptide library in an animal.

Phase II: We will partner with pharmaceutical companies and choose the best phase I technologies to identify the indication, libraries, and methods to attempt treating a real disease.

Why Now? | First, the drug discovery problem needs to be addressed as soon as possible. The drug development cost doubled every 9 years and just recently stopped at ~\$1B per approved drug¹⁶. Care for rare incurable diseases costs \$768B/year¹⁷, yet curing them one at a time is economically viable¹⁸.

Second, there is a growing interest in challenging the current paradigm in drug discovery¹⁶, making it ripe for a paradigm shift¹⁹. For example, the National Institute of Health recently proposed new alternative methods (NAMs) due to low translation success of animal models. FDA, established the exploratory Investigational new drug (eIND) pathway in 2006 which allows a part of drug discovery to be done in a early clinical trials²⁰, e.g. via microdosing to measure the pharmacokinetics of the new drugs in humans²¹.

Third, necessary technologies just became available. They just need to be focused and integrated. Recently, researchers realized that at microdose it can be possible to show localized therapeutic effects²², and that several such drugs can be delivered at once to evaluate effects on the diseased tissue²³ **in humans**²⁴. Finally, the gene editing technologies, a recent Nobel prize for directed evolution, and industrial efforts for screening AAVs popularized the use of in vivo high-throughput screening technologies^{6,25-27}. Peptide drugs are also a versatile class of therapeutics with >80 of such drugs already on the market²⁰. Our synthetic serum markers can enable noninvasive testing of drug action delicate tissues^{12-15,28,29}.

Key Stakeholders and performers: We plan to involve several groups. First, academic labs in in vivo pooled screening and histological analysis. Second, disease experts to identify phenotypic criteria for screening. Third, drug delivery experts for library and drug candidate delivery. We will also include pharma companies to aid in silico drug toxicity and biodistribution evaluation to improve the input libraries. Lastly, we will include the large-animal facilities to facilitate the progression to pre-IND testing.

Team and Roles: **1)** Jerzy Szablowski Ph.D. is a recently tenured Assistant Professor in Bioengineering at Rice working on highly innovative molecular neurotechnologies. He received a number of national awards including the DARPA YFA, NIH Director's New Innovator, the Packard Fellowship, and Sloan Fellowship. He is a co-founder and CEO of Imprint Bio.; **2)** Technical support, x2 PhD/MD scientists with pharma and clinical translation experience; **3)** Administrative Support x1 and project management

Impact and outcomes | This 5-year program advances drug discovery, culminating in discovery of drug candidates for a currently unsolved disease. We will establish the theoretical basis for in vivo drug screening, create new delivery systems, and enable design of safe drug libraries. Furthermore, the program offers broad systemic advantages and risk mitigations. Even in the absence of drug success, real human disease data will provide ground-truth training for better AI models. Localized delivery unlocks massive-scale screening of drug combinations, addressing multi-target diseases is currently problematic.

Budget | To ensure a critical path is successful, we aim for 24 projects ranging from \$350-500k (for computational studies), through \$1.5-2.5m small animal and molecular biology studies, and \$3-4m large animal studies, for a total of \$30m in Phase I, and \$5-10m for pre-IND studies in Phase II for one indication.

References

- 1 Miller, G. (American Association for the Advancement of Science, 2010).
- 2 Thomas, D. *et al.* (2021).
- 3 Schober, A. Classic toxin-induced animal models of Parkinson's disease: 6-OHDA and MPTP. *Cell and tissue research* **318**, 215-224 (2004).
- 4 Barabási, A.-L., Gulbahce, N. & Loscalzo, J. Network medicine: a network-based approach to human disease. *Nature reviews genetics* **12**, 56-68 (2011).
- 5 He, B. *et al.* Combination therapeutics in complex diseases. *Journal of cellular and molecular medicine* **20**, 2231-2240 (2016).
- 6 Li, H. R. *et al.* Engineering viral vectors for acoustically targeted gene delivery. *Nature Communications* **15**, 4924 (2024).
- 7 Maturana, C. J., Verpeut, J. L., Kooshkbaghi, M. & Engel, E. A. Novel tool to quantify with single-cell resolution the number of incoming AAV genomes co-expressed in the mouse nervous system. *Gene therapy* **30**, 463-468 (2023).
- 8 Upright, N. A. *et al.* Behavioral effect of chemogenetic inhibition is directly related to receptor transduction levels in rhesus monkeys. *Journal of Neuroscience* **38**, 7969-7975 (2018).
- 9 Waring, M. J. *et al.* An analysis of the attrition of drug candidates from four major pharmaceutical companies. *Nature reviews Drug discovery* **14**, 475-486 (2015).
- 10 Olson, H. *et al.* Concordance of the toxicity of pharmaceuticals in humans and in animals. *Regulatory toxicology and pharmacology* **32**, 56-67 (2000).
- 11 Van Norman, G. A. Limitations of animal studies for predicting toxicity in clinical trials: is it time to rethink our current approach? *JACC: Basic to Translational Science* **4**, 845-854 (2019).
- 12 Lee, S. *et al.* Engineered serum markers for non-invasive monitoring of gene expression in the brain. *Nature Biotechnology*, 1-9 (2024).
- 13 Lee, S. *et al.* Synthetic serum markers enable noninvasive monitoring of gene expression in primate brains. *Neuron* (2026).
- 14 Nouraein, S. *et al.* Erasable serum markers. *Proceedings of the National Academy of Sciences* **122**, e2511741122 (2025).
- 15 Watanabe, S. *et al.* Monitoring in vivo transcription with synthetic serum markers. *bioRxiv*, 2024.2012.2010.627810 (2024).
- 16 Ringel, M. S., Scannell, J. W., Baedeker, M. & Schulze, U. Breaking Eroom's law. *Nat Rev Drug Discov* **19**, 833-834 (2020).
- 17 Garrison, S. *et al.* The economic burden of rare diseases: quantifying the sizeable collective burden and offering solutions. *Health Affairs Forefront* (2022).
- 18 Crooke, S. T. & Lo, A. W. Establishing a commercial solution for extremely rare genetic diseases. *Nature Biotechnology*, 1-3 (2026).
- 19 Kuhn, T. S. *The structure of scientific revolutions*. Vol. 962 (University of Chicago press Chicago, 1997).
- 20 Rossino, G. *et al.* Peptides as therapeutic agents: challenges and opportunities in the green transition era. *Molecules* **28**, 7165 (2023).
- 21 Burt, T. *et al.* Microdosing and other phase 0 clinical trials: facilitating translation in drug development. *Clinical and translational science* **9**, 74 (2016).
- 22 Burt, T. *et al.* Intra-target microdosing—a novel drug development approach: proof of concept, safety, and feasibility study in humans. *Clinical and Translational Science* **10**, 351-359 (2017).
- 23 Klinghoffer, R. A. *et al.* A technology platform to assess multiple cancer agents simultaneously within a patient's tumor. *Science translational medicine* **7**, 284ra258-284ra258 (2015).
- 24 Gundle, K. *et al.* (BMJ Publishing Group Ltd, 2024).
- 25 Chow, R. D. & Chen, S. Cancer CRISPR screens in vivo. *Trends in cancer* **4**, 349-358 (2018).
- 26 Chow, R. D. *et al.* AAV-mediated direct in vivo CRISPR screen identifies functional suppressors in glioblastoma. *Nature neuroscience* **20**, 1329-1341 (2017).
- 27 Zheng, X., Thompson, P. C., White, C. M. & Jin, X. Massively parallel in vivo Perturb-seq screening. *Nature protocols* **20**, 1733-1767 (2025).
- 28 Buitrago, N., Brau, J. & Szablowski, J. O. Modeling synthetic serum marker kinetics for monitoring deep-tissue gene expression. *bioRxiv*, 2025.2011.2017.688787 (2025).
- 29 Seo, J. P. *et al.* Acoustically targeted measurement of transgene expression in the brain. *Science Advances* **10**, ead7686 (2024).

